Quiz 2 - Solutions

Name: Student ID:

Problem 1. Out of 5 men and 7 women, a committee consisting of 2 men and 3 women is to be formed. In how many ways can this be done if
(a) any men and any women can be included?
(b) one particular women must be on the committee?
(c) two particular men cannot be on the committee?

5 men	7 women

(a) $\underbrace{\binom{5}{2}}_{M}\underbrace{\binom{7}{3}}_{W}$

(b) Now 1 woman cannot be on the committee, hence hence we have to choose the 3 women from the rest of 6 women.

$\underbrace{\binom{5}{2}}_{M}\ \underbrace{\binom{6}{2}}_{W}$

(c) Similar to part (b), Now two men must not be on the committee, hence, 2 men must be choose from the rest of 3 men left.

$\underbrace{\binom{3}{2}}_{M}\underbrace{\binom{7}{3}}_{W}$

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Problem 2. Assume we have 4 boys and 5 girls. We want to arrange them to sit in a row.
a. What is the probability that the boys and the girls are each to sit together?
b. What is the probability if only the boys must sit together?

(2 a) E = The event that the girls and the boys must seat together.

$$P(E) = \frac{4!\,5!\,2!}{9!}$$

(2b) F = The event that only boys must seat together

$$P(F) = \frac{6!\,4!}{9!}$$

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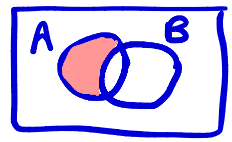
Problem 3. Suppose $P(A) = 0.4$, $P(B') = 0.3$, and $P(A \cap B') = 0.1$.
(a) Find $P(A \cup B)$.
(b) Find $P(B \cap A')$.

(a) $P(A) - P(A \cap B') = P(A \cap B)$



$A \cap B'$

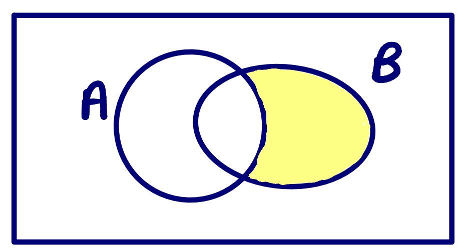
$P(A \cap B) = 0.4 - 0.1 = 0.3$

$P(B) = 1 - P(B') = 1 - 0.3 = 0.7$

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.4 + 0.7 - 0.3$

$P(A \cup B) = 0.8$

(b) $P(B \cap A') = P(B) - P(B \cap A) = 0.7 - 0.3 = 0.4$



$B \cap A'$